

Arjun Giriraj Day 08

TASK 1 – NETWORK IDENTITY (8 marks)

From ipconfig /all on YOUR computer record:

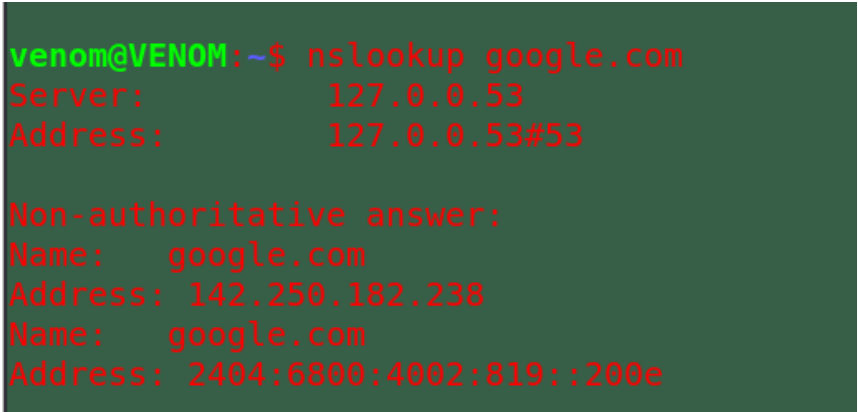
- IPv4 Address: 10.179.195.85
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.179.195.87
- DNS Server: 10.179.195.87 AND 2401:4900:87a2:e6fa::f
- MAC/Physical Address: 10:6f:d9:7a:da:a9
- DHCP Enabled (Yes/No): Yes

TASK 4 – DNS INVESTIGATION (6 marks)

Run nslookup [google.com](https://www.google.com)

State:

- which DNS server responded: 127.0.0.53
- which IP address was returned: 142.250.182.238
- and what the purpose of DNS is: DNS translates the human readable name such as [google.com](https://www.google.com) into IP addresses so computer can communicate.
- Attach a screenshot.



```
venom@VENOM:~$ nslookup google.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.182.238
Name:   google.com
Address: 2404:6800:4002:819::200e
```

TASK 5 – NETWORK PATH INVESTIGATION (6 marks)

Run traceroute google.com. State:

- how many visible hops appeared: 18
- what the first hop was: 2401:4900:87a2:e6fa::f
- what device commonly represents the first hop: local router/default gateway
- what traceroute is used for: To trace the network path between source and destination.
- Attach a screenshot.

```
venom@VENOM:~$ traceroute google.com
traceroute to google.com (2404:6800:4013:811::8b), 30 hops max, 80 byte packets
 1  2401:4900:87a2:e6fa::f (2401:4900:87a2:e6fa::f)  4.017 ms  3.991 ms  3.985 ms
 2  * * *
 3  fc00:0:2:73::1 (fc00:0:2:73::1)  206.099 ms * *
 4  * * *
 5  2404:a800:1a00:805::131 (2404:a800:1a00:805::131)  206.067 ms  206.061 ms  2404:a800:1a00:805::135 (2404:a800:1a00:805::135)  206.041 ms
 6  * * *
 7  2001:4860:1:1::14d4 (2001:4860:1:1::14d4)  200.582 ms * *
 8  2001:4860:0:1::7819 (2001:4860:0:1::7819)  200.451 ms 2001:4860:0:1::78d3 (2001:4860:0:1::78d3)  200.446 ms 2001:4860:0:1::7819 (2001:4860:0:1::7819)  166.709 ms
 9  2001:4860:0:1::77d8 (2001:4860:0:1::77d8)  166.694 ms 166.689 ms 241.900 ms
10  2001:4860:0:1::8d19 (2001:4860:0:1::8d19)  241.894 ms 2001:4860:0:1::8d25 (2001:4860:0:1::8d25)  241.889 ms 2001:4860:0:1::8d19 (2001:4860:0:1::8d19)  25.173 ms
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * lcdely-in-fl39.1e100.net (2404:6800:4013:811::8b)  76.593 ms
```